Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. For $A(1;2)$ and $B(4;-2)$, calculate $\overrightarrow{AB}$.
2. Calculate the norm of $(3;-4)$.
3. Simplify $\overrightarrow{AB}+\overrightarrow{BC}$.
4. Expand $(a+b)^2$.
5. In a right triangle, an angle has adjacent side 3 and hypotenuse 5. Give its cosine.
6. What is the midpoint of $A(-2;1)$ and $B(4;3)$?

After the diagnostic. Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

Speaking and logic. Compare, on the same problem, coordinates and vector calculus without a frame. Justify why your method uses the data most directly.

Investigate before formalising

Construct three points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for a segment AB of 6 cm. Conjecture the locus, then produce a proof that includes A and B.

Organisation. Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.